Low-Rank Structure Is Geometry

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Beginner worksheet · 30–40 minutes · no proofs or code from scratch

Open the named Pluto Lab, use only the relevant experiment control, inspect the visual result, and answer briefly. The interactive version is `notebooks/05_ExerciseSheet.jl`. The themes are adapted at an introductory level from Paul Breiding's Tensorlab exercises; the questions are rederived for TensorKitchen and are not verbatim copies.

The goal is to distinguish the represented object, its coordinates, the model assumption, and the evidence needed before interpreting components.

Exercise 1 — Keep the axes or flatten them?

notebooks/00_Primer.jl · Section 1 · about 6 minutes

Compare a sample × token × feature activation tensor of size (3, 4, 5) with the matrix obtained by flattening sample and token together.

1. How many entries are in the 3 × 4 × 5 tensor?

□ 12 □ 20 □ 60 □ 120

2. If sample and token are flattened into one axis, what is the matrix size?

□ (3, 20) □ (12, 5) □ (4, 15) □ (60, 1)

3. Does flattening delete any numerical entries?

□ Yes □ No

4. What information is no longer explicit in the 12 × 5 matrix?

5. Which view is more useful for comparing token patterns across samples, and why?

Exercise 2 — Build, decompose, reconstruct

notebooks/00_Primer.jl · Section 3 · about 7 minutes

Generate a rank-3 tensor from A of size (3, 3), B of size (3, 3), and C of size (2, 3). Fit a rank-3 CPD and inspect the returned object.

1. What is the size of the generated tensor, and how many rank-one terms were used?

2. What is the format of the returned CP decomposition?

3. Reconstruct T-hat. What size does it have?

4. Record ||T - T-hat||F / ||T||F from the notebook.

5. If the relative error is nearly zero, what has been verified?

6. What has not been verified by a near-zero error?

7. Must the fitted factor columns equal the original columns entry by entry?

☐ Yes ☐ No

Exercise 3 — Same object, different coordinates

notebooks/01_OneObjectManyCoordinates.jl · Sections 1A–1B · about 8 minutes

Open the Matrix gauge experiment. Make a prediction, then move the gauge slider and compare coordinate change with object change.

Measure

1. Set $\log10(s) = 0$. Record s and $\kappa(Q)$.

2. Set $\log10(s) = 3$. Record s and $\kappa(Q)$.

3. Tick every quantity that changes strongly.

□ factor coordinates □ factor conditioning □ represented matrix

4. In the CP rescaling and permutation experiment, does the represented tensor change?

□ Yes □ No

Predict and explain

5. For $Q(s) = \mathrm{diag}(s, s^{-1})$, what do you expect as s becomes very large?

□ X changes dramatically □ the factors may become badly scaled

□ the reconstruction error stays near zero □ $\kappa(Q)$ increases

6. If $A' = AQ$ and $B' = BQ^{-T}$, complete $A'B'^{T}$ = _____.

7. Two different factor pairs produce the same matrix. Which object should we regard as the underlying object?

□ A □ B □ the pair (A, B) □ the represented matrix X

8. If $X = A_1B_1^T = A_2B_2^T$, must $A_1 = A_2$ and $B_1 = B_2$?

□ Yes □ No

9. A relative reconstruction error is 10^{-14}. Which conclusion is justified?

□ the factors are unique □ the recovered matrix is extremely close

□ the problem is well-conditioned □ every factor was recovered correctly

10. Finish the sentence: A low-rank factorization is a _____ of the underlying matrix. The coordinates can change while the represented _____ stays the same.

Exercise 4 — Mode subspaces and model choice

notebooks/02_GeometryAtlas.jl · Model cards and capacity experiment · about 8 minutes

Compare CP, Tucker, and BTD on the two-block target. Focus first on the structure each model preserves; unfolding-rank arithmetic is an optional extension.

Read the structural assumption

1. In a Tucker model, must every mode use the same compression rank?

□ Yes □ No

2. What does the Tucker core describe?

3. Which object is a CP component: one mode vector or the complete outer product across all modes?

□ One mode vector □ The complete outer product

4. Why should raw Tucker factor columns not automatically be treated as unique concepts?

Choose the structural model

5. Which model uses one shared list of rank-one components across all modes?

□ CP □ Tucker □ BTD

6. Which model uses one core with a possibly different compression rank for each mode?

□ CP □ Tucker □ BTD

7. The Lab 2 target is a sum of two small Tucker blocks. Which model most directly mirrors that construction?

□ CP □ Tucker □ BTD

8. If one fitted model has the smallest error in this finite run, does that prove it is always the best model?

□ Yes □ No

Exercise 5 — Observe stagnation, diagnose its geometry

notebooks/03_OptimizationFailureMuseum.jl · about 10 minutes

Use the failure map, component-similarity control, three-case comparison, solver race, and plateau microscope to separate an observation from its possible cause.

Collision and reliable ALS updates

1. Move the control from Distinct toward Almost identical. What happens to component separation?

□ It decreases □ It stays fixed □ It increases

2. Do the two complete rank-one patterns become easier or harder to distinguish?

□ Easier □ Harder

3. As component separation decreases, what happens to ALS update sensitivity?

□ It decreases □ It stays fixed □ It increases

4. In plain language, what becomes hard for ALS near a collision?

5. Press Redistribute shared signal near a collision. What changes strongly, and what changes very little?

Trajectory and plateau diagnosis

6. Why must every solver in the race start from the same CPDPoint?

7. What observation triggers the plateau flag, and which diagnostics are used afterward to test a collision explanation?

8. The poor-start case is slow even though its components remain separated and update sensitivity is moderate. Does slow optimization by itself prove collision?

□ Yes □ No

9. Can a reconstruction error be small while individual components remain difficult to interpret reliably?

□ Yes □ No

Exercise 6 — From activations to a concept claim

notebooks/04_NeuralRepresentations.jl · Concept explorer and evidence ladder · about 14 minutes

Audit one candidate concept in the synthetic CRAFT-style pipeline. Separate what the factorization shows from the semantic and behavioral claims that still require evidence.

Read the factorization

1. What does one row of the activation matrix A represent?

2. In $A \approx UW^T$, what do U[i,j] and W[:,j] represent?

3. Move the rank slider from 1 to 5. At which rank are the three planted ingredients easiest to name, and what happens on either side?

4. Choose one concept. Which evidence helps you propose a human label: the CAV alone or the high-usage crops?

□ CAV alone □ High-usage crops

Audit the claim

5. In the perturbation experiment, is the most present concept necessarily the most important to the class score?

□ Yes □ No

6. Why is nonnegativity not enough for a concept claim? Give one stability check.

7. Why is the sample × location × feature factorization in 4H not CRAFT?